



Language Model

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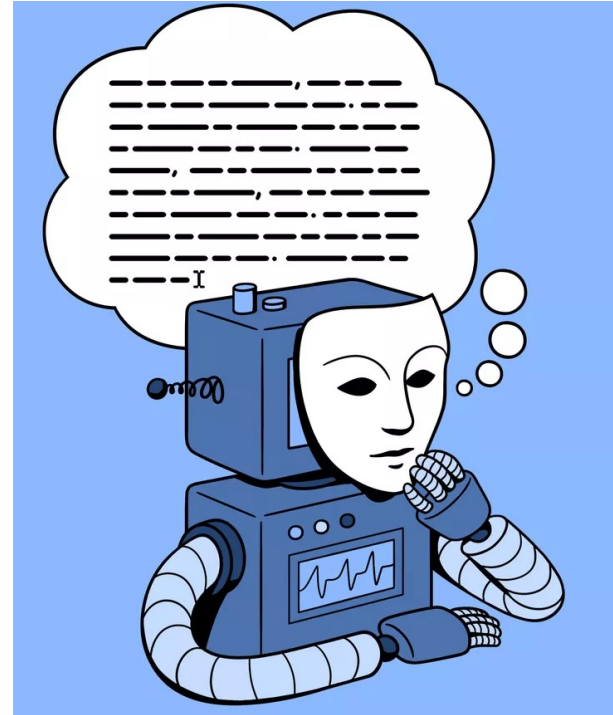
Preceptor



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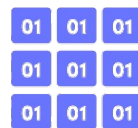
Objectives

- Vectorizing Unstructured Data
- Bag-of-Word Models
- N gram representation
- Probabilistic Language Modeling
- Estimating n-gram Probabilities
- Evaluation



Learning from Unstructured data

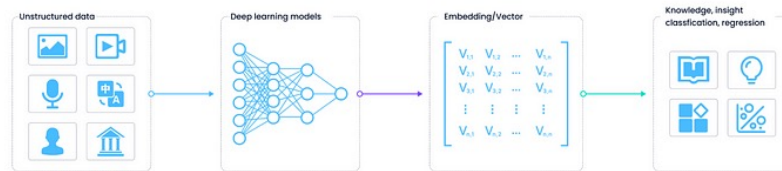
- Structured data is ideal for Machine Learning
 - Databases with tabular data populated with numerical features
- 80% of available data in unstructured. ML algorithms cannot directly work with them
 - Text data in the form of speech or written medium
 - Videos or images



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01	01	01

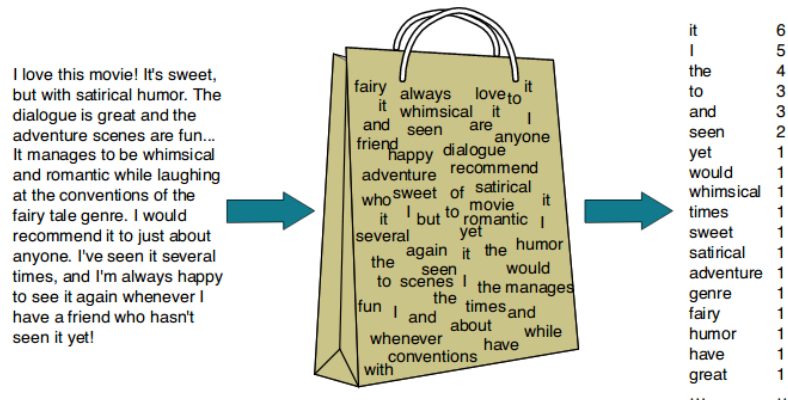


- **Text Vectorization**
 - Words and phrases in open text act as features
 - Need a process post pre-processing to transform their presence/absence into a numerical representation of text data → Vectors/Tensors
- Start with setting a vocabulary, V : set of all the possible distinct words encountered in a text corpus.
- We explore strategies that convert text data into $|V|$ dimensional vectors of binary or real-valued features.



Bag-of-Words representations

- Text Vectorization technique: View text as an unordered collection of words. Corpus is represented as a term-document matrix
- One-hot Encoding:** Treat each word in vocabulary as a categorical variable
 - Indicator variable indicating presence or absence of word in text.
 - Downside: Frequency counts can be important in measuring relative importance of words within corpus.
- Count Vectorization:** Replace indicator variables with raw counts of word frequencies.
- TF-IDF Model:** Improvement over the previously described BOW model that simply records word counts
 - Term Frequency – $tf(w, D)$ represents the frequency or raw count of word w in document d .
 - Inverse Document Frequency – Log transform of word w 's document frequency (df). $idf(w, D) = \log \frac{N}{df(w)}$
 - $tfidf(w, D) = tf(w, D) \times idf(w, D)$

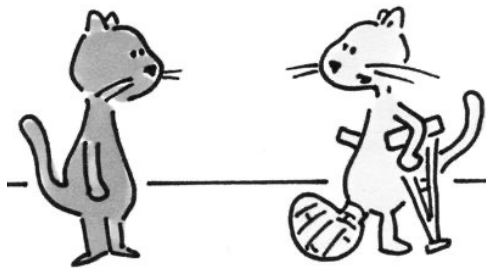


N-gram representation

- Bag-of-words models only include lexical features that do not capture meaning or context.
- N-grams: Sequence of n consecutive words/tokens within text
 - Contain meaning better than individual words (that are called *unigrams*)
 - Example: N-grams for *The cat killed curiosity* →
 - Bigrams:** *The cat; cat killed; killed curiosity*
 - Trigrams:** *The cat killed; cat killed curiosity*
 - We can train models to learn the likelihood (probability) of n-grams in a text corpus to predict more coherent phrases/sentences : Most NLP tasks are formulated in this manner.
 - N-grams are the cornerstone of language generation and understanding.
- Models that assign probabilities to upcoming words, or sequences of words in general, are called language models
- **Language models can assign a probability to an entire sentence.**
 - *all of a sudden I notice three guys standing on the sidewalk* gets a much higher probability in well-trained language models than *on guys all I of notice sidewalk three a sudden standing the*

The cat killed curiosity.
Curiosity killed the cat.

Opposite sense but same BOW representation



"It's a good thing I was only mildly curious!"

N-gram probabilities

- We want to calculate the probability of a word w given some history h (*sequence of words that precede it*)
 - Example: its water is so transparent that → what do you think is the most likely word to follow?
 - $P(\text{the} | \text{its water is so transparent that})$
- One way to estimate this probability is from relative frequency counts.
 - Taking a very large corpus count the number of times
 - we find *its water is so transparent that* in the corpus
 - we find *its water is so transparent that the* we see in the corpus
 - $$P(\text{the} | \text{its water is so transparent that}) = \frac{\text{Count}(\text{its water is so transparent that the})}{\text{Count}(\text{its water is so transparent that})}$$
- Downside:
 - Sparsity Issues - many ways to create sentences hence very specific ones will always be rare still
 - Counting is not scalable - If we wanted to know the joint probability of the entire phrase *its water is so transparent*, we do it by asking “out of all possible sequences of five words, how many of them are *its water is so transparent*?” That’s a lot to estimate!



N-gram probabilities

- We want to find clever ways to approximate these probability of a word w given a history h
- For a sequence of n words represented either as $w_1 \dots w_n$ or $w_{1:n}$ we want to compute
 - Joint Probability: $P(w_1, w_2, \dots, w_n)$ i.e. $P(w_{1:n})$
 - Conditional Probability: $P(w_n | w_1, w_2, \dots, w_{n-1})$
- **Chain Rule Approximation:**
$$\begin{aligned} P(w_1, w_2, \dots, w_n) &= P(w_n | w_1, w_2, \dots, w_{n-1}) P(w_1, w_2, \dots, w_{n-1}) \\ &= P(w_1) P(w_2 | w_1) P(w_3 | w_1, w_2) \dots P(w_n | w_1, w_2, \dots, w_{n-1}) \\ &= \prod_{k=1}^n P(X_k | X_{1:k-1}) \end{aligned}$$
- We can obtain the joint probability for a sentence by multiplying the conditional probabilities but how do we deal with counting and sparsity? → **Markov assumption**
- Intuition of n-gram model: instead of computing the probability of a word given its entire history, we can approximate the history by just the last $n - 1$ words. For example,
 - In a **bigram** model $P(\text{the} | \text{its water is so transparent that}) \approx P(\text{the} | \text{that})$
 - In a **trigram** model $P(\text{the} | \text{its water is so transparent that}) \approx P(\text{the} | \text{transparent that})$
- Applying this Markov assumption for a bigram model in the previous equation for joint probability we get, $P(w_{1:n}) \approx \prod_{k=1}^n P(w_k | w_{k-1})$



Estimating n-gram probabilities

- How do we estimate these bigram or n-gram probabilities? - Maximum Likelihood Estimate
 - To compute the bigram probability of a word w_n given the previous word w_{n-1} we'll compute the following
 - Count of the bigram $C(w_{n-1}w_n)$
 - Count of all bigrams with the first word $w_{n-1} : \sum_w C(w_{n-1}w)$
 - $P(w_n|w_{n-1}) = \frac{C(w_{n-1}w_n)}{\sum_w C(w_{n-1}w)}$
 - Next, we use the chain rule to compute the joint probability of a sentence under the Markov assumption for bigrams.
 - Since floats obtained for probabilities can cause underflow issues we often prefer to represent and compute them in log format.

Example: We have a mini corpus of three sentences. To begin and end sentence generation we use two special symbols:

1) $\langle s \rangle$ at the start of sentence and 2) $\langle /s \rangle$ at the end of sentence.

$\langle s \rangle$ I am Sam $\langle /s \rangle$

$\langle s \rangle$ Sam I am $\langle /s \rangle$

$\langle s \rangle$ I do not like green eggs and ham $\langle /s \rangle$

Here are the calculations for some of the bigram probabilities from this corpus

$$P(I|\langle s \rangle) = \frac{2}{3} = .67 \quad P(\text{Sam}|\langle s \rangle) = \frac{1}{3} = .33 \quad P(\text{am}|I) = \frac{2}{3} = .67$$

$$P(\langle /s \rangle|\text{Sam}) = \frac{1}{2} = 0.5 \quad P(\text{Sam}|\text{am}) = \frac{1}{2} = .5 \quad P(\text{do}|I) = \frac{1}{3} = .33$$



Evaluating Language Models



- Requirements : 1) Training Set 2) Development Set and 3) Test Set
- Since language models are simply learned probability distributions their evaluation is not very straightforward.
- **Extrinsic Evaluation:** Often n-gram language models are a component of a separate downstream NLP task like speech recognition or sentiment analysis. The quality of a LM can be assessed by the effectiveness of the resulting NLP engine at the end task.
- **Intrinsic Evaluation:** Sometimes end-to-end evaluation is expensive because it needs additional dataset collection, labeling, training, and evaluation.
Alternative measure: **Perplexity**
 - Inverse probability of a test sentence (text), normalized by the number of words.
$$\text{perplexity}(W) = P(w_1 w_2 \dots w_N)^{-\frac{1}{N}}$$
Applying chain rule of probability, we can rewrite the previous equation as
$$\text{perplexity}(W) = \sqrt[N]{\prod_{i=1}^N \frac{1}{P(w_i | w_1 \dots w_{i-1})}}$$
You can apply further n-gram specific Markov approximations to simplify the previous term.
 - Perplexity is a measure of surprise, and we want to diminish that in probabilistic modeling. Lower the perplexity, the better your language model is at predicting words.

N-gram Models: Concluding Thoughts

- Sampling sentences from a Language Model - Sampling from a language model—which represents a distribution over sentences—means to generate sentences according to their likelihood, i.e., joint probability of tokens, as defined by the model.
- Generalization and Zeros – N-gram models are dependent on training corpus for the quality of their probability estimates.
 - Contextually relevant corpora are very important.
 - N-grams do a better job of modeling as we increase the value of N
 - Sparsity – Some zero-probability n-grams should have non-zero probabilities assigned by the LM for them to be produced at test time when appropriate.
 - Zero probabilities lead to underestimation of specific phrases/n-grams
 - Also, if the probability of any word in the test set is 0 in training the entire probability of a test sentence containing it becomes zero.
 - Solution: Smoothing algorithms that shave off a bit of probability mass from some more frequent events and give it to these unseen events.
 - Unknown Words - What do we do about words we have never seen before?
 - Closed Vocabulary with sub-word tokenization
 - Open Vocabulary to model unknown words in the test using <UNK> token

1 gram	-To him swallowed confess hear both. Which. Of save on trail for are ay device and rote life have
2 gram	-Hill he late speaks; or! a more to leg less first you enter -Why dost stand forth thy canopy, forsooth; he is this palpable hit the King Henry. Live king. Follow.
3 gram	-What means, sir. I confess she? then all sorts, he is trim, captain. -Fly, and will rid me these news of price. Therefore the sadness of parting, as they say, 'tis done. -This shall forbid it should be branded, if renown made it empty.
4 gram	-King Henry. What! I will go seek the traitor Gloucester. Exeunt some of the watch. A great banquet serv'd in; -It cannot be but so.

**Sentences randomly generated from LMs
trained from Shakespeare and WSJ articles**

1 gram	Months the my and issue of year foreign new exchange's september were recession exchange new endorsed a acquire to six executives
2 gram	Last December through the way to preserve the Hudson corporation N. B. E. C. Taylor would seem to complete the major central planners one point five percent of U. S. E. has already old M. X. corporation of living on information such as more frequently fishing to keep her
3 gram	They also point to ninety nine point six billion dollars from two hundred four oh six three percent of the rates of interest stores as Mexico and Brazil on market conditions